Question:
If the sum of all the angles except one of a convex polygon is 2190°, then the number of sides of the polygon must be
$\textbf{(A)}\ 13 \qquad \textbf{(B)}\ 15 \qquad \textbf{(C)}\ 17 \qquad \textbf{(D)}\ 19 \qquad \textbf{(E)}\ 21$

Reasoning + Answer:

<think>
1. **Understanding the problem**: We are given that the sum of all the angles except one of a convex polygon is \(2190^\circ\). We need to find the number of sides of the polygon.

2. **Sum of interior angles of a polygon**: The sum of the interior angles of a polygon with \(n\) sides is given by the formula:
\[
(n-2) \times 180^\circ
\]

3. **Given information**: The sum of all the angles except one is \(2190^\circ\). Let \(S\) be the sum of all the angles in the polygon. Therefore, the angle that is missing must be \(S - 2190^\circ\).

4. **Relating the missing angle to the polygon**: Since the polygon is convex, each interior angle is less than \(180^\circ\). Let \(x\) be the missing angle. Then, the sum of the remaining angles is:
\[
S - x
\]

5. **Setting up the equation**: We know that the sum of the remaining angles is \(2190^\circ\). Therefore:
\[
S - x = 2190^\circ
\]

6. **Finding \(x\)**: Since \(S\) is the sum of all the interior angles of the polygon, we can substitute \(S\) with \((n-2) \times 180^\circ\):
\[
x = (n-2) \times 180^\circ - 2190^\circ
\]

7. **Condition for convex polygon**: For the polygon to be convex, each interior angle must be less than \(180^\circ\). Therefore, the missing angle \(x\) must be such that the remaining angles still sum up correctly.

8. **Solving for \(n\)**: We need to find \(n\) such that the sum of the remaining angles is \(2190^\circ\). We can use the fact that the sum of the interior angles of a convex polygon is less than \(180^\circ \times n\). Therefore, we need to find \(n\) such that:
\[
(n-2) \times 180^\circ - 2190^\circ < 180^\circ \times n
\]

9. **Simplifying the inequality**:
\[
180n - 360 - 2190 < 180n
\]
\[
-360 - 2190 < 0
\]
\[
-2550 < 0
\]
This inequality is always true, so we need to find the specific \(n\) that satisfies the condition.

10. **Finding \(n\)**: We know that the sum of the remaining angles is \(2190^\circ\). We need to find \(n\) such that:
\[
(n-2) \times 180^\circ - 2190^\circ = 180^\circ \times k \text{ for some integer } k
\]
Solving for \(n\):
\[
180n - 360 - 2190 = 180k
\]
\[
180n - 2550 = 180k
\]
\[
n = k + 14
\]

11. **Checking the options**: We need to check the options given:
- \(n = 13\): Sum of interior angles = \(13 \times 180 - 219